

Put all this together, and you have cloud computing, which is a very broad term that covers a range of resources and services offered over the Internet. Some experts opine that any resource that a company uses over the Internet, outside its firewall, is 'on the cloud'. So, a company can choose, customise or develop an enterprise-wide solution, manage and maintain it, scale it up or down, or do whatever it wants, completely over the Internet—oh, sorry, we should now be saying "on the cloud", but it means the same thing anyway. The cloud is nothing but a metaphor for the Internet, if that helps ease the confusion in any way!

...the cat and the fiddle

The biggest advantage of cloud computing is that the end-user or developer does not have to bother about the physical location or configuration of the actual resources. The service providers will worry about all that, abstracting all the dirty details from the developers, who can work completely on a logical plane.

These systems can be easily scaled up or scaled down. You can start with a small server and a little storage during development, and then scale up the volume or features of the system on-the-go, as the usage grows. Or, a company can use cloud resources simply to cater to temporary needs. They could even opt to use a combination of their own and cloud resources. In short, they can use what they want, and pay just for what they use.

Almost all literature on this subject compares the cloud to the electricity grid. You consume electricity without bothering about where it is generated, how far away or along which route it travelled to get to you. Nor do such details as how many others are using power from that grid, bother you. Similarly, you can use software, services and infrastructure from the service providers using a Web interface, without worrying about any back-end details. Examples of cloud-computing platforms include Amazon's Elastic Cloud Compute (EC2), Salesforce's Force.com and Microsoft's Azure.

The abstraction provided by cloud computing is so beneficial, from a user's perspective, that some large organisations are cloaking even their existing infrastructure as 'private clouds'. What this means is that they consolidate all their resources and deliver it to various user units *within* the company, just as cloud service providers would do to the public. So, the users within the company can use the resources they need without worrying about where they are, or how they are managed. To the company, it offers a two-fold benefit—user convenience, as well as better resource utilisation. No resource gets locked up. It is all on the private cloud, can be managed centrally, and provisioned as needed within the company. Plus, the life of resources also gets extended. An old, slow and steady computer can still be used, as long as it can connect to the network—it can run all the needed applications off the cloud!

"While the benefits of cloud computing are compelling, as with any product or technology, adoption takes its own course. Indian organisations, like many of their counterparts across the world, are in evaluation mode. They have largely moved away from the understanding mode, but adoption is not yet high. I think India presents an immense opportunity to realise the cloud possibilities. In developed countries, we see organisations looking at cloud adoption in stages: first, adopting infrastructure virtualisation; then, setting up a private cloud; this is followed by selective use of the public cloud; before moving a large part of the application portfolio to the cloud (private or public). For Indian organisations, there is a big opportunity for them to leapfrog to the cloud faster, given their under-investment in information technology (IT). So they don't have to worry about managing legacy investments, creating an expensive migration plan, and then adopting the cloud," says Dhiraj Sinha, leader of the Applications Technology group, Dell Services. Many of Dell's solutions have used open source—for example, Dell is working with Canonical to help customers adopt Ubuntu-powered open source clouds, and the high-performance analytics services of Dell use Hadoop, the open source distributed computing and data-storage framework.

"Open source has built the cloud"

Sharing, the freedom to mix and match, choice, and many other characteristics of cloud computing demonstrate a likeness to the principles of open source software. More than just the similarities, there is also a sheer necessity for open source software and open standards in clouds that comprise heterogeneous, and often proprietary, infrastructure. The availability of source code, the freedom to modify and redistribute, the flexibility and constant evolution, and other open philosophies greatly favour the cloud schema of things. A *cloud* is, after all, a fluffy and lovable structure. Would we not hate a boxed cloud?

Undoubtedly, open source lies at the foundation of many of the earliest cloud implementations. "Open source has built the cloud. When we think of the services we consume on the cloud, from Facebook and Google to Amazon, none would have been affordable or scalable using a traditional licensing model," says Prakash Advani, partner manager—Central Asia, Ubuntu. "Moving that capability from the leading edge of SaaS provision and into the mainstream enterprise is the next big opportunity for open source."

The fact that most of the public clouds, including Amazon's Elastic Compute Cloud (EC2), run on Linux-based platforms is just one facet of the story. In fact, that is now taken so much for granted that the focus has shifted to tools and platforms that enable the building of private or hybrid clouds, the integration of legacy infrastructure with the cloud, and so on.